

Week 12

1. Sockets (I – Networking)

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Content

- 1 Introduction to Computer Networks
- 2 Network models
 - OSI Model
 - TCP/IP Model
- 3 Client/Server
- IP Addresses

Part I

Introduction to Computer Networks



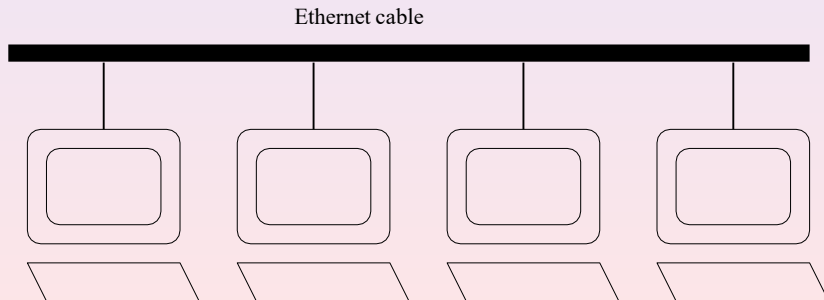
Introduction to computer networks

Computers: evolution

- **The old model:** A single computer serving users who bring their works for processing → became obsolete.
- **The new model:** A large number of separate but interconnected computers are serving users → **computer networks.**

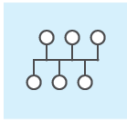
Computer network

- **Interconnected computers:** they are able of exchanging information
- **Computer network:** interconnected collection of autonomous computers.



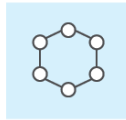
Networking Topologies

6 types of network topology



Bus

Directly connects devices to each other and transmits data between links.



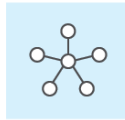
Ring

Connects devices next to each other in the form of a circle. Communication occurs unidirectionally or bidirectionally.



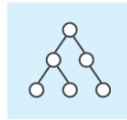
Mesh

Connects each device to every other device in the network.



Star

Features a central device which transmits data to other nodes in the system.



Tree

Connects devices down in a structure resembling a tree where parent nodes connect to child nodes.



Hybrid

Consists of at least two different types of network topology.

Some typical network applications:

Applications

- Exchange electronic mail (e-mail)
- Exchange files
- Remote login
- Share peripheral devices
- Execute a program on another computer

IPC: Review

IPC Methods

- Using Files
- Using Memory
- Unnamed Pipes: Related processes
- Named Pipes (FIFO): Unrelated processes

Network IPC: Sockets

Sockets

- IPC between processes running on different computers

Part II

Network Layering: OSI



Layering

Given a specific task, say file transfer, it is hard to solve the entire problem in one step. Solving this problem needs:

- A user interface
- Coding data into packets
- Observing some protocols
- Routing the packets
- A physical data transmission

Layers

Networks are organized in **layers**, where:

- The number of layers and their functions differ from network to network
- Each layer offers some services to the higher layer.
- A same level 'layer peer' on different machines can converse using a **protocol**: rules and conventions used in this conversation.

Network models

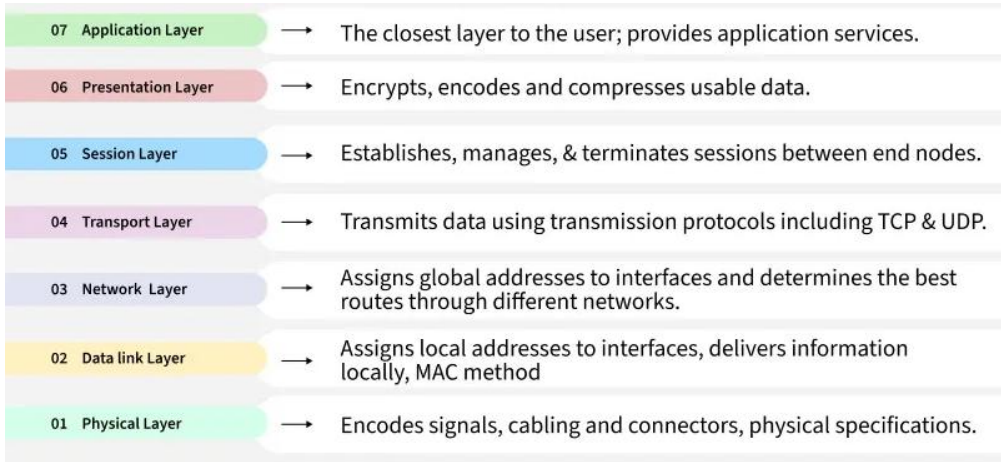
Computer Network Model describes how information is transmitted between two points in a network. Two models:

- The **OSI model** (Open Systems Interconnect) proposed by ISO describes network activities as having a structure of **seven layers**, each of the layers has one or more protocols associated with it. → Theoretically sound.
- The **TCP/IP model** (Transmission Control Protocol/Internet Protocol) based on the TCP/IP protocol suite. → **More practical.**

○ The OSI Reference Model



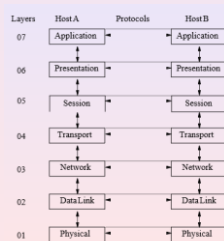
The OSI Reference Model



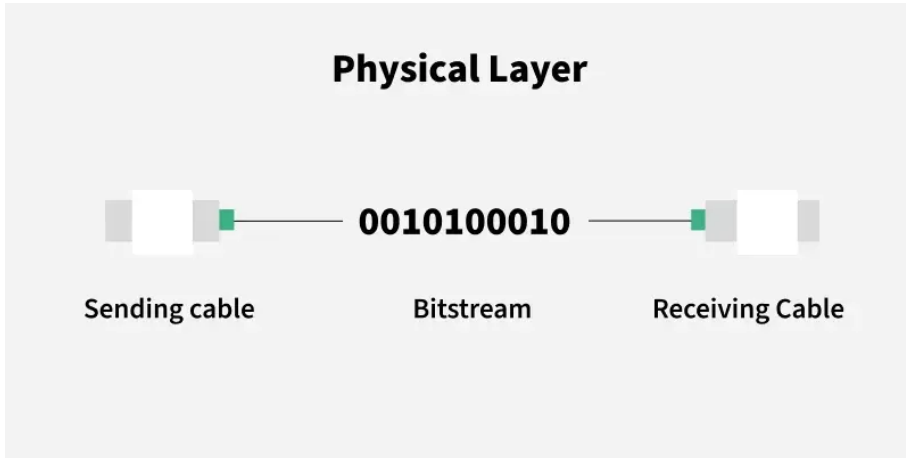
The OSI Reference Model

Each layer defines a function performed when data is transferred between applications across a network.

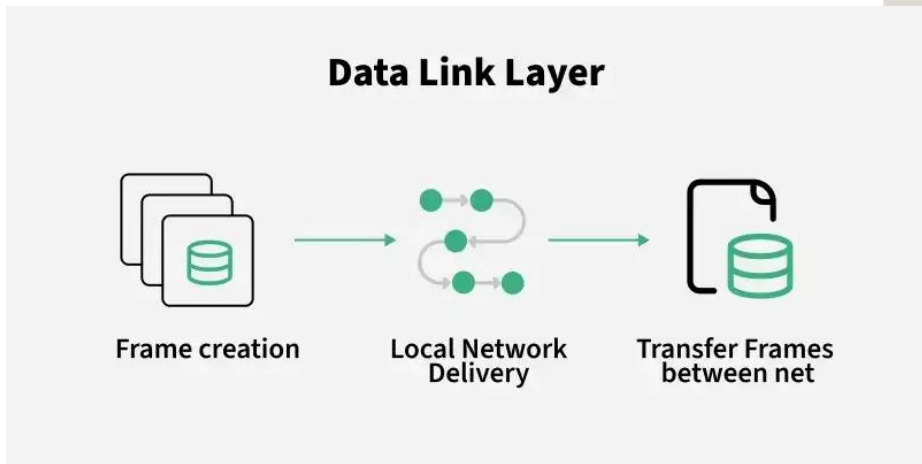
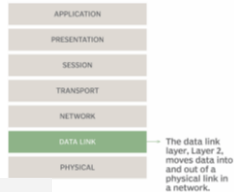
1. **Application Layer:** application programs that use the network.
2. **Presentation Layer:** standardizes data presented to the applications.
3. **Session Layer:** manages sessions between applications.
4. **Transport Layer:** provides error detection and correction.
5. **Network Layer:** manages network connections.
6. **Data Link Layer:** provides data delivery across the physical connection.
7. **Physical Layer:** defines the physical network media. Consists of cable and connectors and is responsible for converting analog signals into bits.



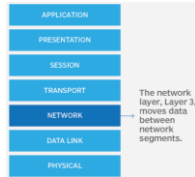
OSI – Layer 1: Physical



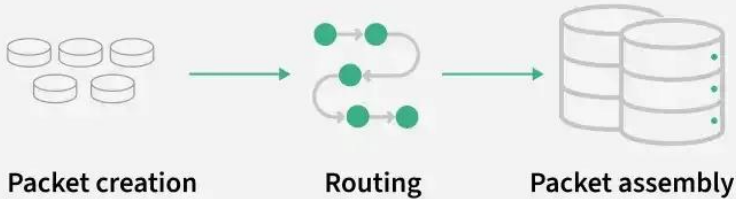
OSI – Layer 2: Data Link



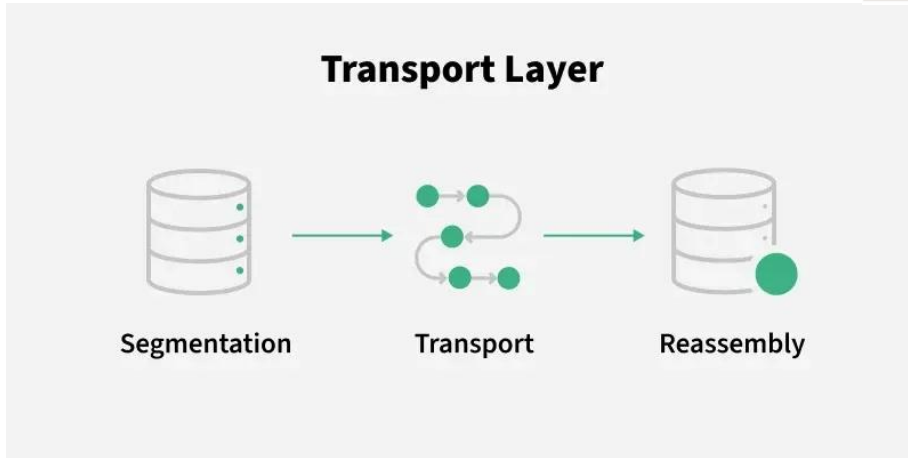
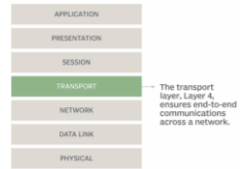
OSI – Layer 3: Network



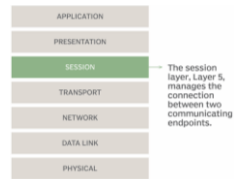
Network Layer



OSI – Layer 4: Transport



OSI – Layer 5: Session Layer

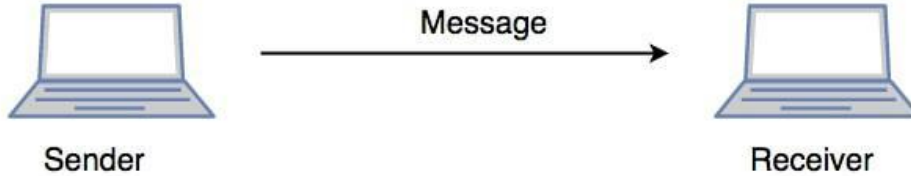
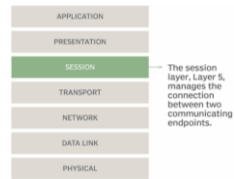


Session Layer

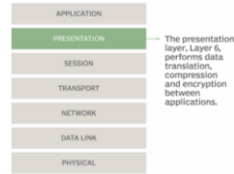


Session of Communication

OSI – Layer 5: Session – Example



OSI – Layer 6: Presentation Layer

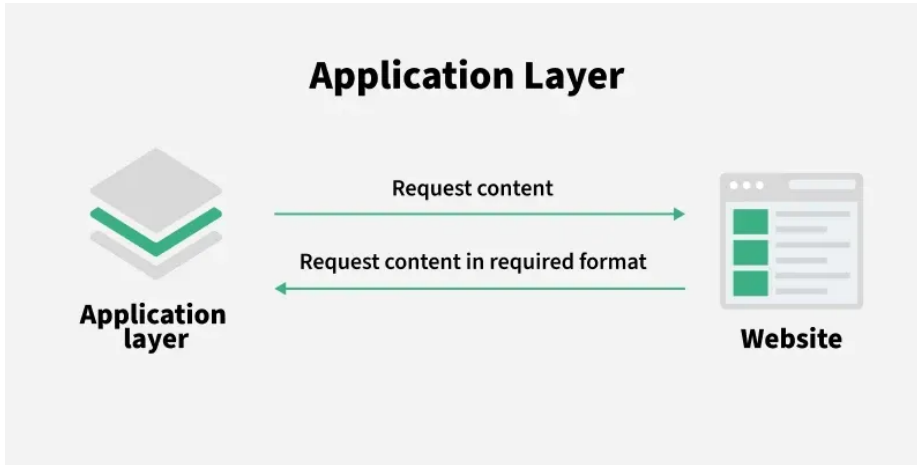


Presentation Layer

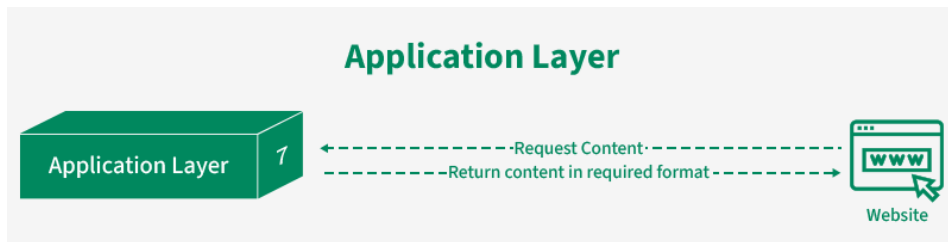
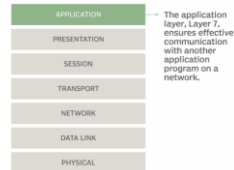


OSI – Layer 7: Application

APPLICATION	→ The application layer, Layer 7, ensures effective communication with another application program on a network.
PRESENTATION	
SESSION	
TRANSPORT	
NETWORK	
DATA LINK	
PHYSICAL	



OSI – Layer 7: Application Flow



○ Protocols used in the OSI Model



Protocols used in the OSI Model

Layer	Working	Protocol Data Unit	Protocols
1: Physical Layer	Establishing Physical Connections between Devices.	Bits	USB , SONET/SDH , etc.
2: Data Link Layer	Node to Node Delivery of Message.	Frames	Ethernet , PPP, etc.
3: Network Layer	Transmission of data from one host to another, located in different networks.	Packets	IP, ICMP , IGMP , OSPF , etc.
4: Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segments (for TCP) or Datagrams (for UDP)	TCP , UDP , SCTP , etc.
5: Session Layer	Establishes Connection, Maintenance, Ensures Authentication and Ensures security.	Data	NetBIOS , RPC , PPTP , etc.
6: Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Data	TLS/SSL , MIME , etc.
7: Application Layer	Helps in identifying the client and synchronizing communication.	Data	FTP , SMTP , DNS , DHCP , etc.



○ Data Flow in the OSI Model

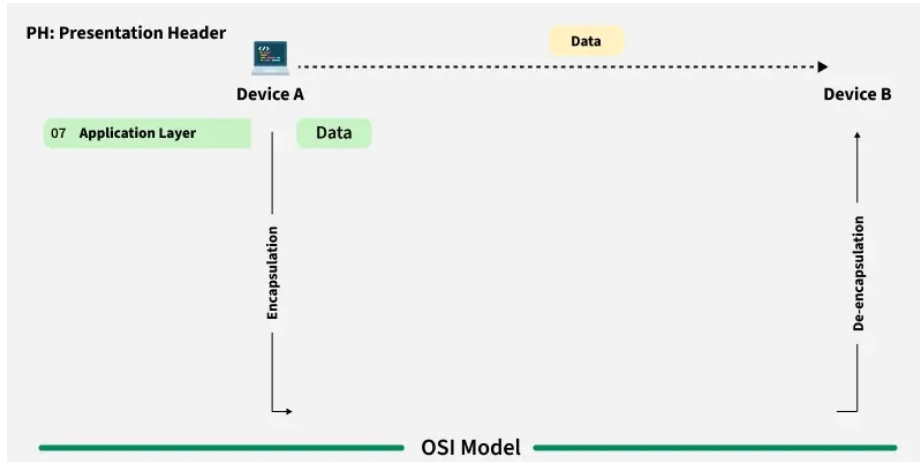


The OSI Reference Model

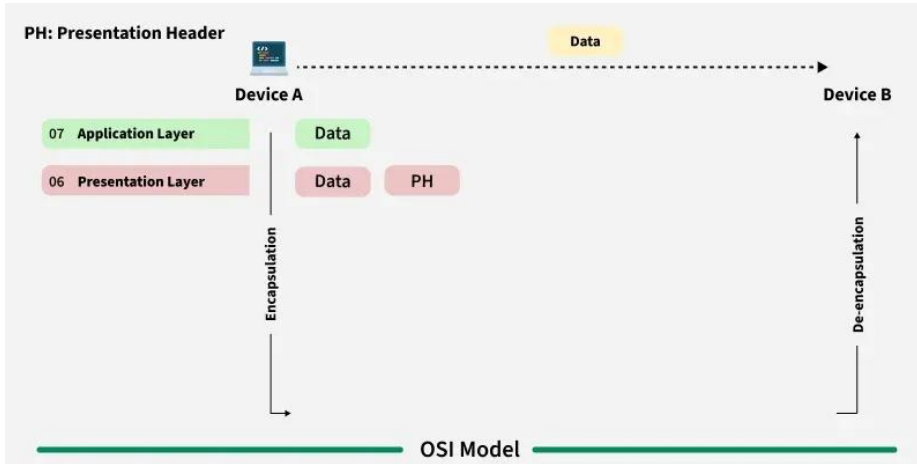
- Information is **passed down** through the layers until it is transmitted across the network, where it **is passed back up** the stack to the application at the remote end.
- Each layer relies on the other layers to perform their functions.
- Individual layers **do not care** how the other layers operate. They only need to know how to pass information up or down from one layer to another.



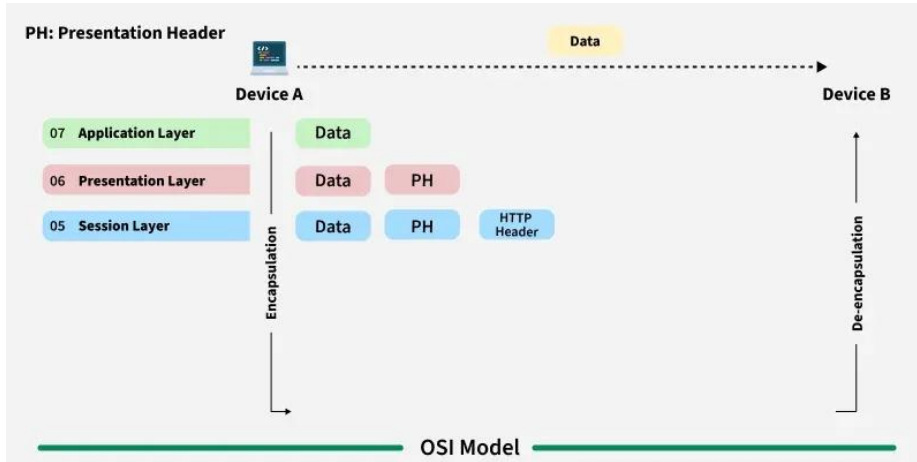
Data Flow in the OSI Model



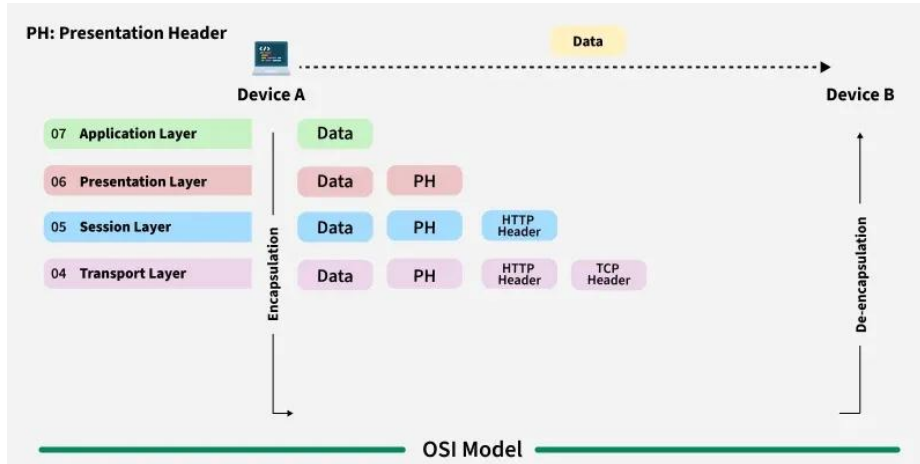
Data Flow in the OSI Model



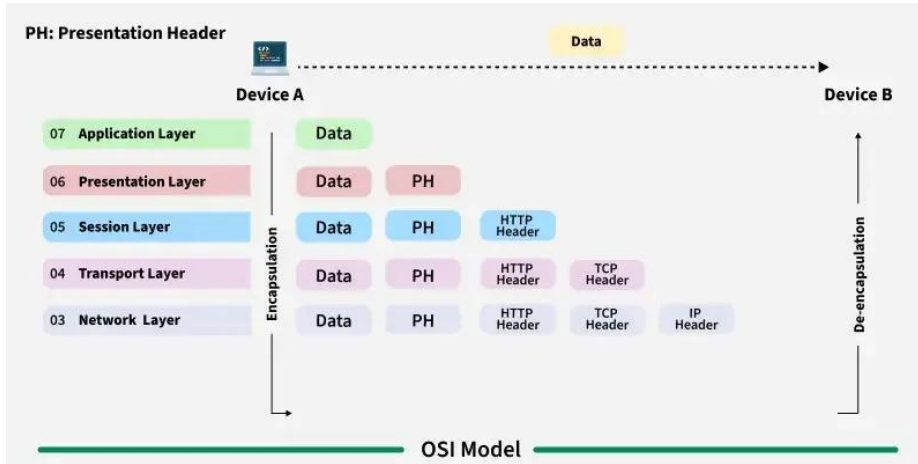
Data Flow in the OSI Model



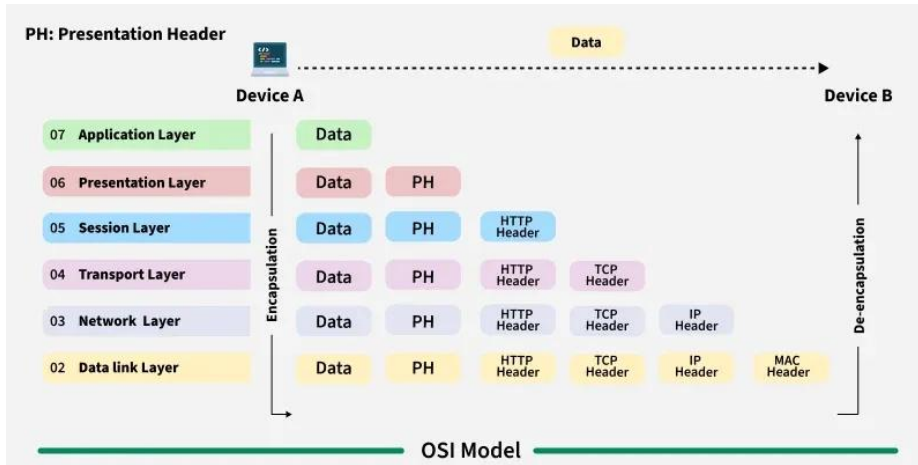
Data Flow in the OSI Model



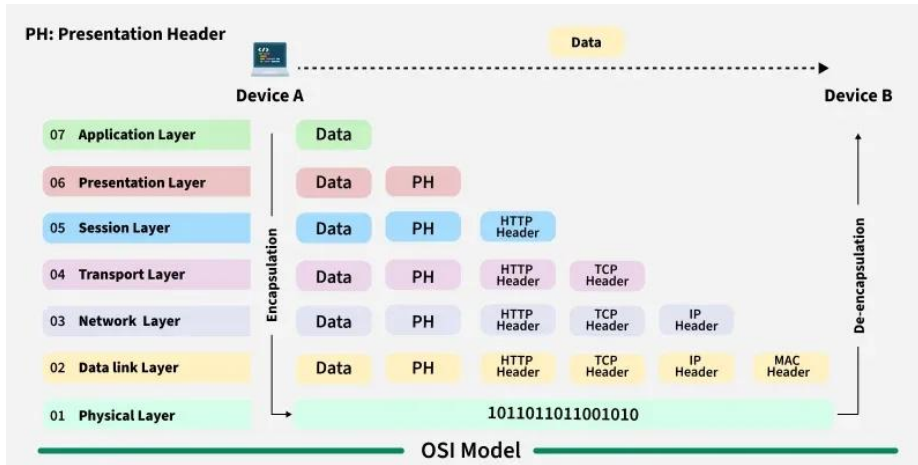
Data Flow in the OSI Model



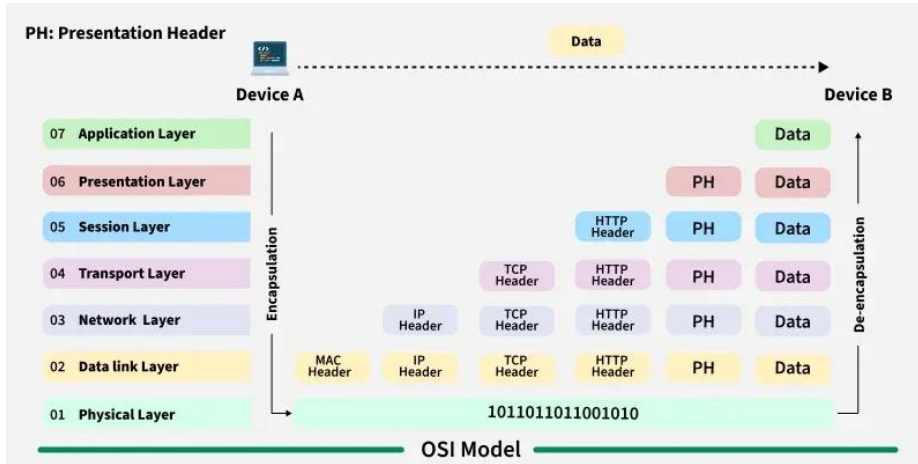
Data Flow in the OSI Model



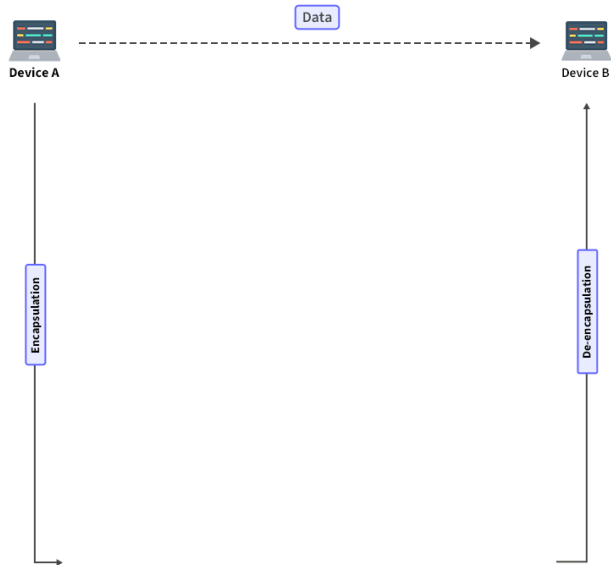
Data Flow in the OSI Model



Data Flow in the OSI Model



Data Flow in the



* PH : Presentation Header



Peer conversation

- In each layer a function may be performed by any number of protocols.
- Each protocol communicates with a peer that is an equivalent implementation of the same protocol on a remote system.
- Each protocol layer is only concerned with communication to a peer at the other end of a link.

Example

- E-mail is an application level protocol that communicates with a peer e-mail application on a remote system,
- FTP is an application level protocol that communicates with a peer ftp application on a remote system.

These applications **do not care** whether the physical layer is a serial modem line or an ethernet connection.

Part II

Network Layering: TCP/IP



○ The TCP/IP Model

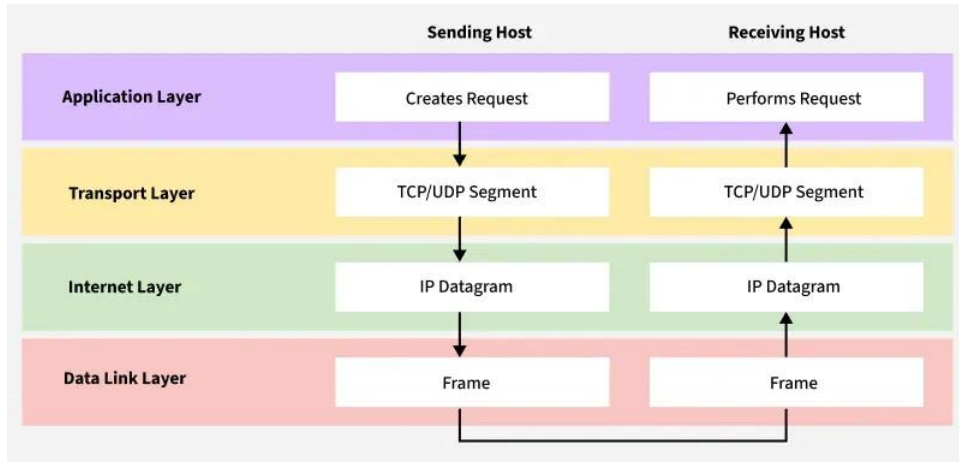


In practice, the OSI Reference Model **is rarely used** when working with networks.

TCP/IP Model

- TCP/IP model is based on the OSI model.
- TCP/IP is usually described **as having fewer layers** than the OSI model.
- TCP/IP does not follow the OSI model exactly.
It is based on the **TCP/IP protocol suite**.
- The documentation on TCP/IP protocol standards is **freely** available and widely published.
- The standards for TCP/IP protocols are developed and maintained independently from any specific hardware type or operating system.
- Widely implemented on most operating systems and computers.
- **TCP/IP is built into the UNIX operating system and is used by the Internet**

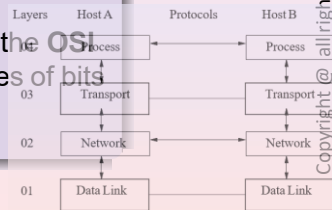
The TCP/IP Model



TCP/IP Model

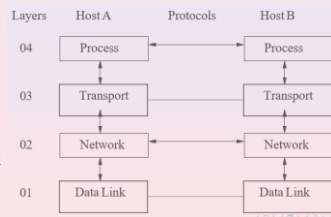
TCP/IP Model

- The **Process Layer**: it combines the top three layers of the **OSI** (application, presentation and session layers) into one. It defines standard Internet services and network applications that user can use (e.g. ftp, telnet...).
- The **Transport layer** transmits messages from a process on one machine to a process on another machine.
- The **Network layer** routes packets through a network to a destination node.
- The **Data Link layer**: it combines the 2 bottom layers of the **OSI model** (Data Link and Physical layers). It transmits frames of bits across a single link between two machines.



Some protocols

- **Process Layer:** Simple Mail Transfer Protocol (**SMTP**) and Post Office Protocol (**POP**) used for e-mail, Hyper Text Transfer Protocol (**HTTP**) used for the World-Wide-Web, and File Transfer Protocol (**FTP**)...etc.
- **Transport Layer:** In TCP/IP there are two Transport Layer protocols:
 - The Transmission Control Protocol (**TCP**) guarantees that information is received as it was sent.
 - The User Datagram Protocol (**UDP**) performs no end-to-end reliability checks.
- **Network Layer:** This is commonly referred to as the Internet Layer. All upper and lower layer communications travel through the Internet Protocol (**IP**) as they are passed through the TCP/IP protocol stack.
- **Data Link Layer:** TCP/IP makes use of existing Data Link and Physical Layer standards rather than defining its own. Data Link Layer describes how IP utilizes existing data link protocols such as **Ethernet**, **Token Ring**,... etc.



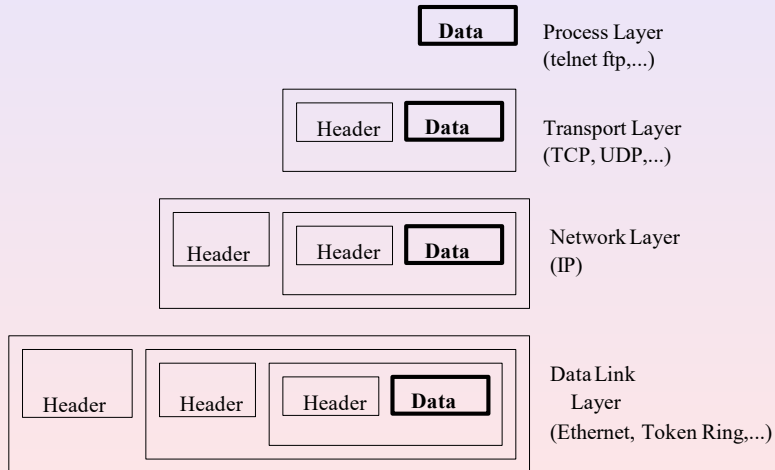
More about TCP/IP protocol

TCP: **T**ransmission **C**ontrol **P**rotocol. A **connection-oriented protocol** that provides a reliable, full-duplex, byte stream for a user process (most used by Internet application).

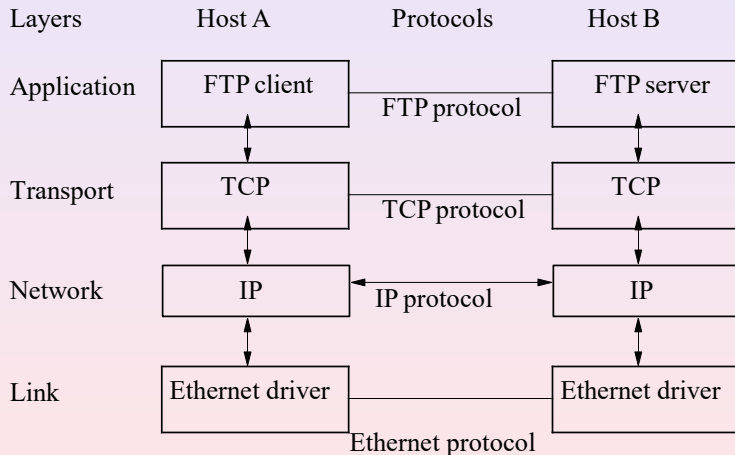
UDP: **U**ser **D**atagram **P**rotocol. A **connectionless protocol** for user processes.

IP: **I**nternet **P**rotocol. IP is the protocol that provides the packet delivery service for TCP and UDP.

Encapsulation



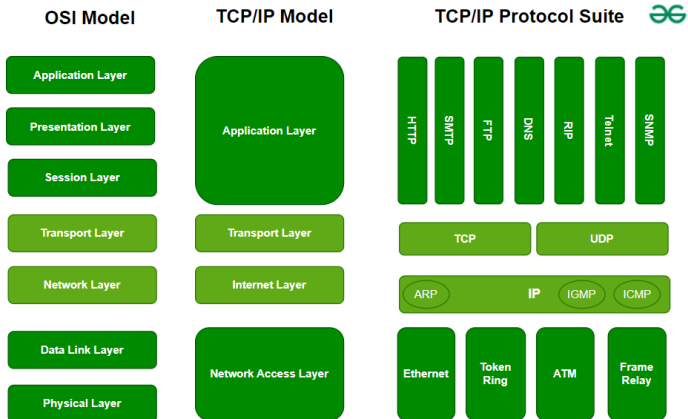
Example



○ TCP/IP vs OSI



TCP/IP vs OSI



TCP/IP vs OSI

Reason	Explanation
Simpler Structure	TCP/IP has only 4 layers, compared to 7 in OSI, making it easier to implement and understand in real systems.
Protocol-Driven Design	TCP/IP was designed based on working protocols, while the OSI model is more of a theoretical framework.
Flexibility and Robustness	TCP/IP adapts well to different hardware and networks and includes error handling, routing, and congestion control.
Open Standard	TCP/IP is open, free to use, and not controlled by any single organization, helping it gain universal acceptance.
Actual Use vs Conceptual Model	The OSI model is great for education and design principles, but TCP/IP is the one actually used in real-world networking.



○ Universal Datagram Protocol (UDP)



The User Datagram Protocol (UDP)

UDP



Use Case :- Video Conferencing



UDP vs TCP

Transmission Control Protocol	User Datagram Protocol
A connection-oriented protocol	A connectionless protocol
The most widely used protocol on the internet	Used for voice over IP, streaming video, gaming and live broadcasts
Guarantees that no packets are missing and all the data that's sent makes it to the intended recipient	Faster and needs fewer resources
Sends packets in order so they can be stitched back together easily	Packets don't necessarily arrive in order
Slower and requires more resources	Allows missing packets, but the sender is unable to know whether a packet has been received
Has a bigger header than UDP, best suited for apps that need high reliability and transmission time is relatively less critical	Better suited for applications that need fast, efficient transmission, such as games



○ Packets and Encapsulation



Packets

- A network packet is a small, structured unit of data that is transmitted across a digital network, like the internet.
- Data is broken down into packets, which are then sent to their destination where they are reassembled to reconstruct the original message.
- Each packet contains a header with routing information, and a payload with the actual data.



Encapsulation

- Encapsulation marks where a packet, or unit of data, begins and ends.
- Decapsulation is the process of removing the header and trailer information from a packet, as it moves toward its destination.

Data header and trailer



Part III

Client/Server Architecture



○ Client/Server Architecture



Client/Server Architecture

- **Client:**

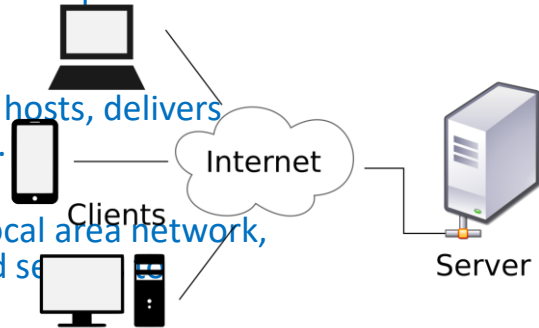
A device or application (e.g., web browser, mobile app, desktop application) that initiates requests for resources or services.

- **Server:**

A powerful computer or system that hosts, delivers and manages resources and services.

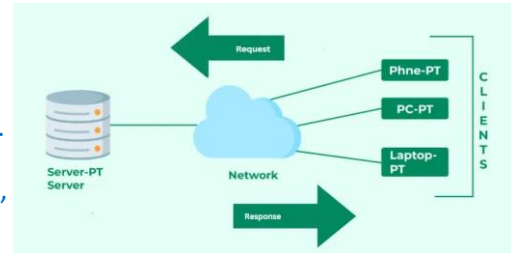
- **Network:**

The communication medium (e.g., local area network, the internet) that enables clients and servers to interact and exchange data.



Client/Server Request-Response Model

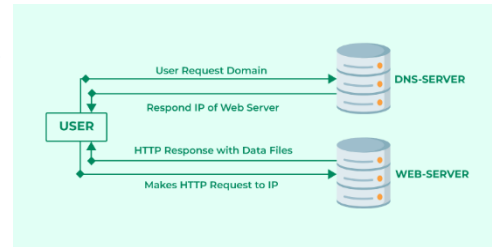
- **Client Request:**
A client sends a request for a specific service or resource to the server over the network.
- **Server Processing:**
The server receives the request, processes it, and performs any necessary actions, such as retrieving data from a database or executing application logic.
- **Server Response:**
The server then sends a response back to the client, containing the requested data or the result of the performed action.
- **Client Processing:**
The client receives the response and processes it, presenting the information to the user or performing further actions based on the server's response.



Client/Server Request-Response Example

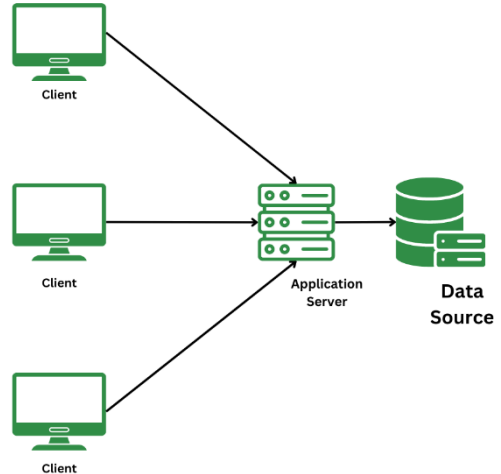
How a Web Browser interacts with the Servers?

- The process of interacting with servers through a browser involves several steps:
 - a. User Enters the URL (Uniform Resource Locator):
The user types a website address (e.g., www.example.com) into the browser's address bar.
 - b. DNS (Domain Name System) Lookup:
The browser contacts a DNS server to convert the domain into an IP address.
 - c. Establishing a Connection:
The browser sends an HTTP/HTTPS request to the server using the resolved IP address.
 - d. Server Responds:
The server sends back website files (HTML, CSS, JavaScript, images).
 - e. Browser Renders the Webpage:
 - a. DOM interpreter: Processes HTML to structure the page.
 - b. CSS interpreter: Applies styles
 - c. JavaScript Engine: Adds interactivity (using JIT compilation for performance).



Client/Server Three-Tier Architecture

- Logical/Physical Tiers:
 - Presentation (Client)
 - Application (Business Logic)
 - Data (Database)
- Advantages:
 - Scalability
 - Modularity and Maintainability
 - Flexibility
 - Security



Client/Server...

Client/Server

The client/server model is the standard model for network applications. A typical scenario is as follows:

- The server process is started on some computer. It initializes itself, then goes to sleep waiting to be contacted by a client process for some service.
- A client process is started on some computer connected to the server's computer with a network. It sends a request across the network to the server asking for some service, remote login for instance.
- The server provides its service in two ways:
 1. For a short amount of time service, the server process handle the service itself and goes back to sleep, waiting for new requests, otherwise
 2. The server invokes another process to handle each client request, so that the server process can go back to sleep.

Part IV

IP Addresses:Ports → Sockets

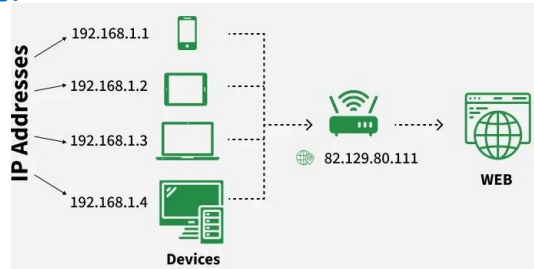


IP Addresses



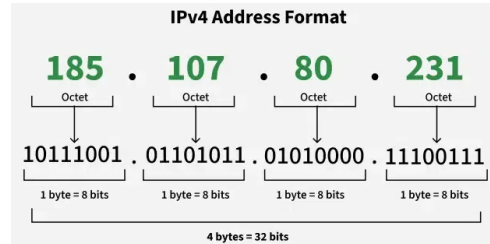
IP Addresses

- A unique numerical label assigned to devices connected to a computer network for communication, like a mailing address.
- Serve two main functions:
 - Identifying a specific device, and
 - Directing data to it.
- There are two main versions:
 - IPv4, and
 - IPv6,and various types, such as public, private, static, and dynamic.



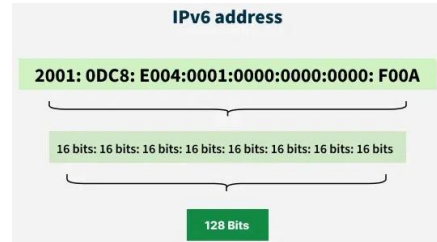
IPv4 Addresses

- Consists of four sets of numbers (octets) separated by dots.
- Supports over 4 billion unique addresses.
- Each octet represents eight bits, or a byte, and can take a value from 0 to 255. This range is derived from the possible combinations of eight bits ($2^8 = 256$ combinations).



IPv6 Addresses

- IPv6 addresses were created to deal with the shortage of IPv4 addresses.
- They use 128 bits instead of 32, offering a vastly greater number of possible addresses.
- These addresses are expressed as eight groups of four Hexadecimal digits, each group representing 16 bits. The groups are separated by colons.





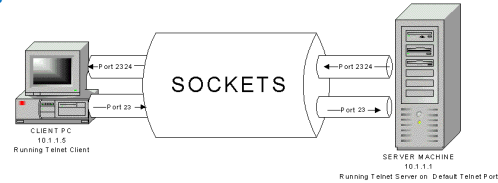
Ports



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Ports

- Logical constructs that identify a specific process or network service on a device.
- Usually a 16-bit number ranging from 1 to 65,535.
- Helps sort incoming traffic to the correct application, preventing a single device from being overwhelmed by multiple services.
- Certain ports are standardized for specific services:
 - Port 80: HTTP (web browsing)
 - Port 443: HTTPS (secure web browsing)
 - Port 25: SMTP (sending email)
 - Port 22: SSH (secure remote access)
- How they work together:
 - An IP address and a port number together form a complete communication endpoint, sometimes called a **socket**.
 - The IP address gets the data to the right computer, and the port number directs it to the correct application on that computer.
- In a URL, the port is often separated from the IP address by a colon, such as in 192.168.1.10:8080.



Part V
Attendance Quiz (Not graded)



Quiz



Attendance Quiz

Thank you!
Questions?

<http://www.linkedin.com/in/andreasmaniatis>



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